

Modelling the impact of climate variability on malaria transmission dynamics across the different ecological zones in Taraba State, Nigeria: A machine learning approach

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ABSTRACT

Background: Malaria transmission in Taraba State, Nigeria, remains a persistent public health challenge, shaped by climatic and ecological factors. Although the influence of weather patterns on malaria incidence has been widely studied, limited research has incorporated ecological zoning into predictive modelling frameworks.

Methods: We analysed malaria incidence data (2021–2023) from the District Health Information Software (DHIS2), alongside rainfall and temperature data from the Nigerian Meteorological Agency (NiMet). Data were cleaned, standardised, and encoded to include ecological and local government area (LGA) variables. Supervised machine learning models, including Decision Tree, Random Forest, and Extreme Gradient Boosting (XGBoost), were trained and validated using four-fold cross-validation. Model performance was assessed with mean absolute error (MAE), mean squared error (MSE), and root mean squared error (RMSE). Shapley Additive Explanations (SHAP) were applied to determine feature importance. Exploratory analyses employed one-way ANOVA to compare malaria incidence across ecological zones and Pearson correlation to test associations with climatic variables.

Results: Exploratory analysis revealed significant heterogeneity in malaria incidence across ecological zones (ANOVA: $F = 5.21$, $p < 0.01$), with the Montane Forest reporting the highest burden, largely driven by Sardauna and Bali LGAs. Rainfall showed a positive correlation with malaria incidence ($r = 0.42$, $p < 0.01$), while temperature was negatively correlated ($r = -0.36$, $p < 0.05$). Across all models, ecological zone consistently emerged as the strongest predictor of malaria incidence, outweighing climatic factors. Random Forest achieved the best predictive accuracy (MAE = 452.6, RMSE = 675.5). SHAP analysis confirmed that geographic variables (zone and LGA) exerted the greatest influence on malaria predictions.

Conclusion: Machine learning approaches may provide a useful framework for exploring malaria transmission dynamics in climate-sensitive and ecologically diverse settings. The findings of this study suggest that ecological zones may be important determinants of malaria incidence in Taraba State. These results indicate that incorporating geographic heterogeneity into malaria surveillance and control strategies could help support more data-driven and locally tailored interventions

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1. Introduction

Malaria remains a significant public health challenge in Nigeria, particularly in Taraba State, where various ecological and climatic factors influence the disease's prevalence.^{1,2} The global phenomenon of climate variability, which encompasses temperature fluctuations, precipitation patterns, relative humidity and wind, further complicates the disease's transmission patterns, affecting not only the human population but also other living organisms.^{3,4} Among these variables, rainfall and temperature are the two most significant drivers of malaria.^{5,6} Between 2030 and 2050, it is anticipated that starvation, malaria, diarrhoea, and heat stress will result in an additional 250,000 deaths annually because of the changes in climatic factors.^{7,8}

Numerous studies have highlighted the impact of temperature and rainfall patterns on mosquito populations and, subsequently, malaria transmission.^{9,10} These climatic factors influence the development of *Plasmodium* parasites within the female *Anopheles* mosquito. The creation and destruction of aquatic habitats, which serve as breeding sites for mosquitoes, are closely tied to rainfall patterns^{11–13}. While seasonal rainfall can facilitate the creation of breeding sites, heavy rainfall can have devastating effects on mosquito populations.^{12,13} Temperature also has a profound impact on mosquito physiology, affecting factors such as longevity, feeding behaviour, and development.^{13,14}

A global analysis of 57 countries from 2000 to 2019 revealed a significant correlation between rising temperatures and increased malaria cases.¹⁵ A separate study utilising an age-structured model demonstrated that malaria transmission is optimal at temperatures between 23.53 °C and 39.80 °C, highlighting the crucial role of temperature in mosquito lifecycle and parasite development.¹⁶ In a regional study conducted in Ghana, researchers found that while the correlation between malaria cases and rainfall intensity was weak across the zones considered, the lag correlations up to four months showed a moderate correlation.¹⁷

While climate plays a significant role in the incidence and spread of malaria, its impact varies across different ecological zones and regions.¹⁸ A study in Taraba State, Nigeria, found that malaria infection rates differ significantly between lowland and highland areas, with 72.6% of the population infected in the lowland compared to 38.6% in the highland.¹⁹ Another study examining malaria transmission dynamics in four ecological zones of Nigeria during the rainy season revealed that people living within the rainforest zone had the highest exposure to female *Anopheles* mosquito bites.¹⁸

Accurate prediction of malaria outbreaks, taking into account ecological zones and climatic factors, can enhance early warning systems and inform resource allocation for malaria control. Incorporating ecological zones and climatic conditions can improve the accuracy of models in predicting malaria outbreaks, enhance early warning systems, and guide more effective allocation of resources for malaria control. Recent advancements in machine learning algorithms offer new opportunities for developing predictive models that incorporate these factors^{20–25}. Studies have demonstrated the effectiveness of machine learning approaches in predicting malaria incidence. For instance, a study in Gambia developed predictive models for malaria outbreaks using various machine learning algorithms, including Extreme Gradient Boosting (XGBoost), decision trees, and support vector machines.⁶ By integrating nine years of historical meteorological and clinical data, the researchers found that XGBoost and decision trees achieved the highest prediction accuracy at 93.3%. The study highlighted the importance of combining climatic and non-climatic factors to enhance the accuracy of malaria outbreak prediction.⁶ Similarly, another study employed partial least squares path modelling and other machine learning algorithms to investigate the relationship between climatic variables and malaria incidence, identifying the Support Vector Machine learning model as the most accurate predictor.²⁶ Furthermore, a study across six Sub-Saharan African countries developed a machine learning-based malaria incidence classification model using climate variability data over 28 years.²⁵

The approach involved feature engineering to identify key climatic factors, k-means clustering for outlier detection, and the XGBoost algorithm for classification. Results showed that non-seasonal changes in precipitation, temperature, and surface radiation significantly influenced malaria outbreaks, with the proposed system outperforming other classification models.²⁵

Although numerous studies have examined the relationship between climatic variables and malaria incidence, few have incorporated ecological zones as part of their predictive modelling. Additionally, to the best of our knowledge, none have identified which of these factors, the climatic variables and the ecological regions, have the strongest influence on malaria incidence in Taraba State. This study addresses this gap by employing a range of supervised machine-learning models to assess the relative importance of these variables in predicting malaria patterns across the state.

2. Materials and method

2.1. Study area

Taraba State, Nigeria, with 16 Local Government Areas (LGAs), has three distinct ecological zones: the Southern Guinea Savanna, Northern Guinea Savanna, and Montane Forest.² Located between 8°00'N and 100°E, the state covers 54,437 km²²⁷ and has a population of approximately 3,493,000, with a malaria prevalence of 24.5%.²⁸ Taraba State has a tropical climate characterised by distinct wet and dry seasons. The wet season spans from April to October, while the dry season persists for at least five months, from November to March.²⁹ Notably, the state's mean annual rainfall exhibits significant spatial variation, ranging from 800 mm in the northern regions to over 2000 mm in the southern

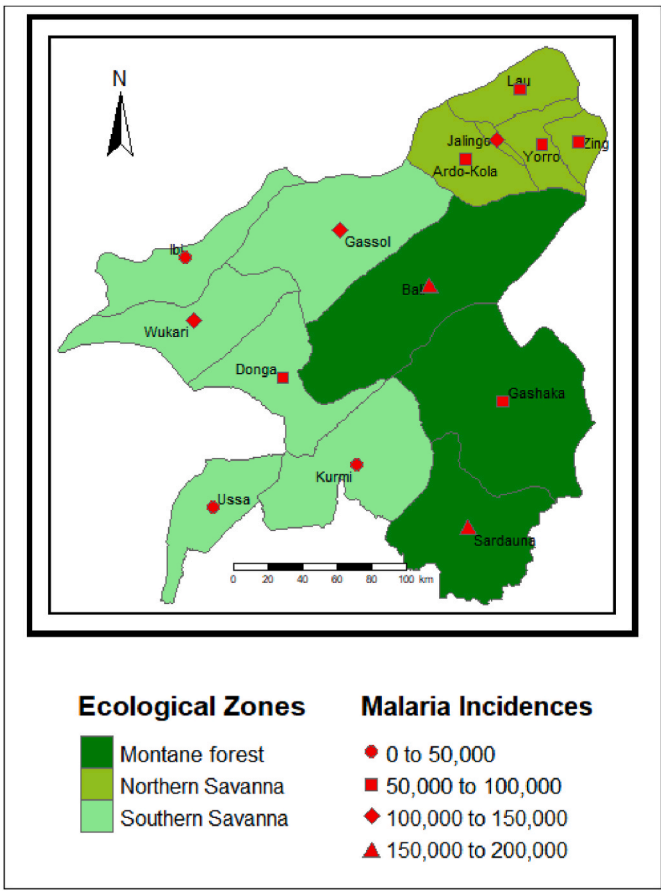


Fig. 1. Map of Taraba State showing the different ecological zones and total malaria incidences for 2021–2023.

areas.²⁹ Fig. 1 displays a map of Taraba State, illustrating the ecological zones, LGAs, and the respective malaria cases.

2.2. Data collection

2.2.1. Malaria data

For this study, secondary data on malaria cases for Taraba state were sourced from the District Health Information System (DHIS2). This platform is used as a repository for health-related data in Nigeria. The data repository was organised to aggregate malaria cases monthly for each LGA within each state. Only microscopy- and rapid diagnostic test (RDT)-confirmed malaria cases occurring from 2021 to 2023 were utilized for the study. The cases include persons with confirmed uncomplicated malaria ≥ 5 years (excluding pregnant women), persons with confirmed uncomplicated malaria < 5 years, pregnant women with confirmed uncomplicated malaria, severe malaria cases seen in persons ≥ 5 years (excluding pregnant women), severe malaria cases seen in persons < 5 years and severe malaria cases seen in pregnant women (PW). These cases were aggregated monthly to calculate the total malaria incidences for each LGA. Although the data used for the study begins in 2021, the effects of the COVID-19 pandemic on healthcare access and service delivery may have persisted beyond 2020. Widespread disruptions to health services were reported in Nigeria during the early phases of the COVID-19 pandemic, which may have had residual impacts on malaria surveillance in the initial months of our study period.³⁰ This observation is corroborated by District Health Information System (DHIS2) metrics on health facility attendance. For instance, in Taraba State, facility visits declined from 1,217,147 in 2020 to 1,081,693 in 2021, representing an 11.1% reduction.

2.2.2. Climate data

Monthly temperature (measured in degrees Celsius) and rainfall data (measured in millimetres) for the study area were acquired from the Nigeria Meteorological Agency (NiMet), covering the same period as the malaria data.

2.3. Machine learning modelling framework

2.3.1. Data pre-processing

To include the zones and the LGAs in the machine learning models used for the study, the zones and LGAs categorical variables were converted into numerical values using the LabelEncoder from the sklearn. Pre-processing module in Python. Since the models considered are not able to handle date-time objects for the date column, this column was converted into an ordinal value to represent time as a single continuous number. The dataset was then chronologically split into training and testing subsets, where 75% of the data was allocated to training and 25% to testing. The split was done temporally to ensure that the model is trained on past data and tested on future data, preserving the natural time order of malaria incidences. The training set was used to train the model, while the test set was reserved for evaluating its predictive performance. The independent variables included temperature, rainfall, time period and categorical encodings for local government area (LGA) and region. The dependent variable was the number of malaria cases.

2.4. Supervised learning models

2.4.1. Extreme Gradient Boosting (XGBoost)

To model the relationship between temperature, rainfall, and malaria incidence, we employed XGBoost, a highly efficient ensemble technique based on gradient boosting, as one of the supervised algorithms.³¹ This approach minimises a cost objective function, J , which comprises a regularisation term and a loss function,³² enabling the identification of complex relationships between climate variables and malaria incidence. The update rule for each iteration can be expressed as³²:

$$\hat{y}_{k+1} = \hat{y}_k - \eta \frac{\partial J}{\partial y_k} \quad (1)$$

For this study, the XGBoost model parameters were tuned using the default parameters except for the learning rate, which was set as 0.01; Number of Estimators: 1000, objective function: regression squared (R^2) error and random state as 42 for reproducibility.

2.4.2. Decision tree regression

A decision tree regressor models decisions and outcomes using a tree-like structure, where internal nodes represent decision rules based on input features, branches correspond to the outcomes of those rules, and leaf nodes indicate the final predicted values.³³ The goal of decision trees is to maximise homogeneity within the splits by recursively partitioning the data, hence reducing variance (or squared error) in each partition for regression problems.³³ For this study, the decision tree model was implemented using the DecisionTreeRegressor from the scikit-learn library. The model was configured with the default parameters, except for the random-state parameter, which was set to 42 to ensure reproducibility.

2.4.3. Random Forest regression (RFR)

RFR works by combining several decision trees during the training phase to produce a more precise outcome than any single node.³⁴ In regression tasks, the output is the average or mean prediction of each tree and due to its robustness and accuracy, the RFR model performs well on a broad range of variables, even those with non-linear correlations.^{34,35} Apart from the Random State set at 42 (to guarantee repeatability) and the Number of Estimators set at 1000 (number of trees in the forest), the model parameters were set up with the default settings for this study.

2.4.4. Model performance evaluation metrics

Mean Square Error (MSE), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE), as presented in equations (2)–(4), were used to evaluate model performance.³⁴

$$MSE = \sum_{i=1}^n (y_i - \hat{y})^2 \quad (2)$$

$$MAE = \frac{\sum_{i=1}^n (y_i - \hat{y})}{n} \quad (3)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y})^2}{n}} \quad (4)$$

Where y_i are the observed values, $\hat{y}$ are the predicted values, and n is the number of data points. To ensure the reliability and generalizability of the results, a fourfold cross-validation approach was applied.³⁶ In addition to these performance metrics, Shapley Additive exPlanations (SHAP) values were computed to provide insights into the contribution of each predictor variable to the model's predictions.³⁷

3. Results

3.1. Exploratory analysis

The following distribution patterns were observed to better understand the relationships between ecological zones, climate, and malaria incidence. Fig. 2 presents the distribution and central tendency of variables across ecological zones. The figure comprises three sub-figures: (a) rainfall distribution (mm), (b) malaria incidence distribution, and (c) temperature distribution ($^{\circ}\text{C}$). The Montane ecological zone displays a bimodal distribution for both rainfall and malaria incidence, with

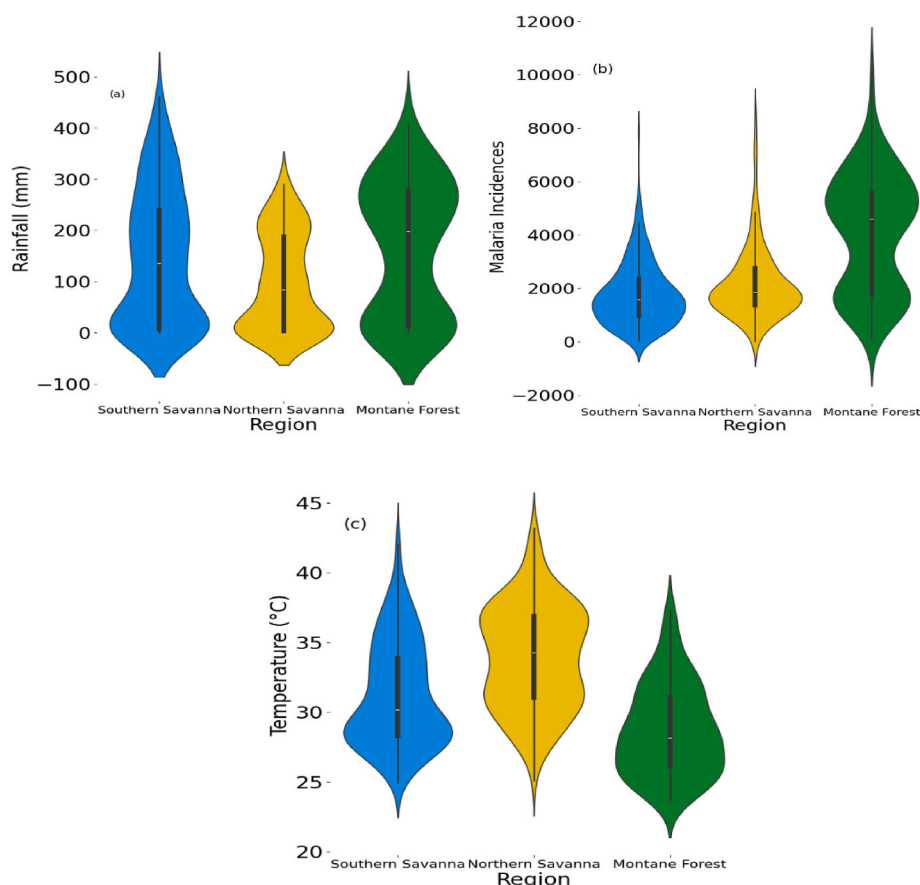


Fig. 2. Distribution of (a) Rainfall (mm), (b) Malaria Incidence, and (c) Temperature (°C) across the ecological zones considered (2021-2023).

median values of approximately 200 mm (Fig. 2a), 5000 cases (Fig. 2b), respectively. These values represent the highest across the zones. This bimodality was further supported by Hartigan's dip test (Dip statistic = 0.083, $p < 0.05$), confirming significant deviation from unimodality. Conversely, the Northern savanna zone exhibits the lowest rainfall, with a median of around 90 mm (Fig. 2a), and the highest temperature, with a median of approximately 34 °C (Fig. 2c). Descriptive statistics for each ecological zone are presented in Table 1, while Comparison of Malaria Incidences, Temperature, and Rainfall Across Ecological Zones is presented in Table 2. A one-way ANOVA test was conducted to statistically verify the observed differences among the zones. The results confirmed that the variations in malaria incidence, temperature, and rainfall were significant ($p < 0.001$).

To identify the Local Government Areas (LGAs) contributing to the high malaria incidence in the Montane Forest, Fig. 3a presents the average malaria incidence across LGAs. Notably, Sardauna and Bali LGAs reported 5440 and 5341 cases, respectively, contributing significantly to the high incidence in the Montane Forest and Taraba State as a whole. Although rainfall and malaria incidence in the Montane Forest

exhibit similar distribution patterns (Fig. 2a and b), a scatter plot (Fig. 3b) reveals no apparent linear relationship between rainfall and malaria incidence. However, high malaria incidence (>90th percentile) is more frequently observed in the Montane Forest with rainfall ranging from approximately 150-350 mm.

3.2. Time-series analysis

Fig. 4 presents the time-series analysis for the ecological zones under study, revealing a direct correlation between malaria incidence and rainfall, and an inverse relationship between malaria incidence and temperature across all the zones. To examine the strength of the correlations across the ecological zones, Fig. 5 illustrates the relationship between malaria incidence and the climatic variables, highlighting variations in correlation strength by zone. From Fig. 5, the Montane Forest has the strongest correlation with rainfall (coefficient: 0.37), while the Northern Savannah has the strongest correlation with temperature (coefficient: -0.49). A seasonal trend in malaria cases was observed across the three zones. In the Northern Savanna, malaria cases

Table 1

Descriptive statistics of malaria incidence, temperature, and rainfall across three ecological zones in Nigeria.

	Southern Savana			Northern Savana			Montane Forest		
	Mal Incidences	Temp. (°C)	Rainfall	Mal Incidences	Temp. (°C)	Rainfall	Mal Incidences	Temp. (°C)	Rainfall
Count	245	245	245	210	210	210	108	108	108
Min	38	24.94	0.87	22	25.09	0.34	46	23.56	0.99
25%	940	28.27	5.98	1363.75	31.03	2.97	1737.25	26.09	11.56
50%	1561	30.16	134.83	1836.5	34.22	83.41	4586.5	28.14	197.38
75%	2383	33.89	239.05	2768.75	36.90	188.10	5573.75	31.11	277.50
Max	7846	42.48	460.65	8532	43.19	290.05	10065	37.26	408.92

Std: Standard deviation; Min: Minimum value; Max: Maximum value; %: percentile.

Table 2
Comparison of malaria incidences, temperature, and rainfall across ecological zones.

Variables	Southern Savanna (Mean ± SD)	Northern Savanna (Mean ± SD)	Montane Forest (Mean ± SD)	p-value
Malaria Incidences	1786.04 ± 1178.19	2182.30 ± 1369.68	4088.70 ± 2173.08	<0.001
Temperature (°C)	31.25 ± 3.77	33.95 ± 3.72	28.69 ± 3.26	<0.001
Rainfall (mm)	146.33 ± 131.60	98.02 ± 92.88	165.06 ± 130.16	<0.001

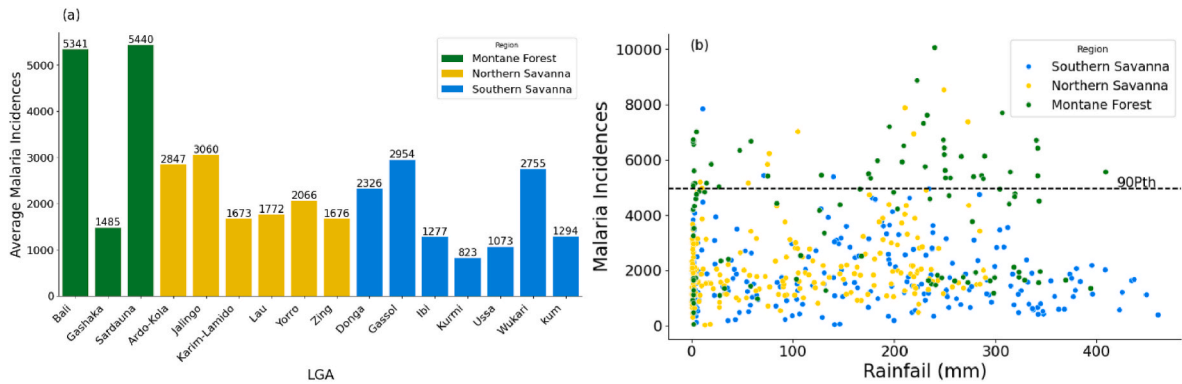


Fig. 3. (a) Average malaria Incidences across LGA (b) Scatter plot of malaria incidence against rainfall (mm).

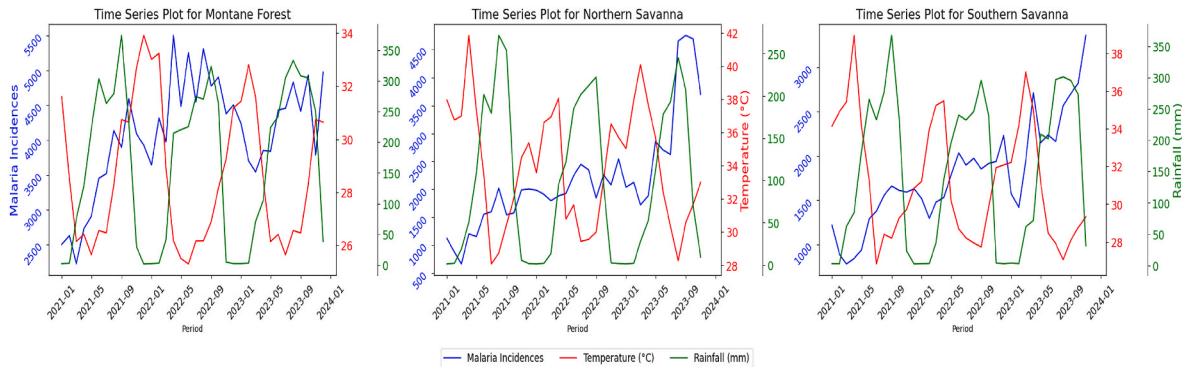


Fig. 4. Temporal analysis of malaria incidence with rainfall and temperature across various ecological zone.

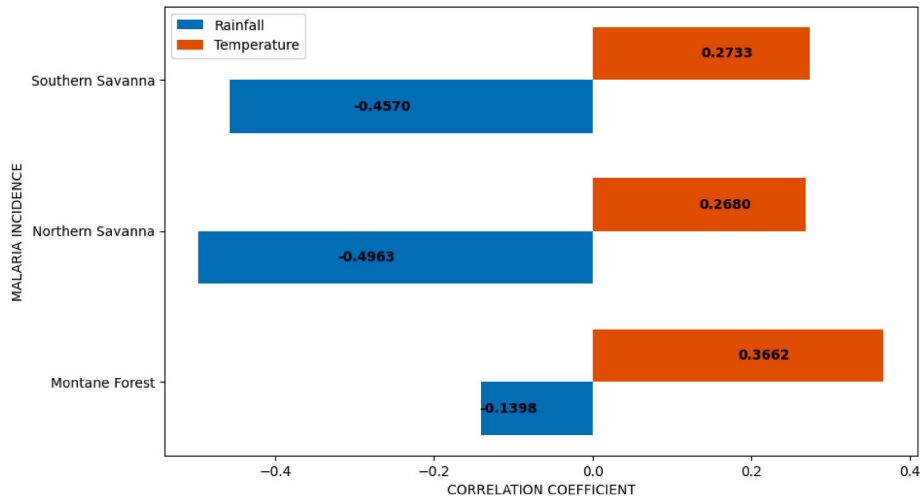


Fig. 5. Correlation between malaria incidence and rainfall, temperature across different ecological Zones.

peaked in August and November 2021, reaching approximately 1700 cases. A similar trend was observed in 2022, with cases rising from 1700 to 2300 between July and August. By 2023, the malaria burden

intensified, with the highest peak of around 4700 cases recorded between October and December. Malaria transmission in the Southern Savanna followed a slightly different pattern. In 2021, the highest

incidence (about 1700 cases) was recorded in September and December. By 2022, the peak shifted to June–July, with cases rising to 2000 and reaching approximately 2200 by November–December. In 2023, the highest peak occurred between November and December, with cases reaching 3,300, while another notable peak of around 2700 cases was observed between April and May. In the Montane Forest region, seasonal peaks were evident, with the highest cases recorded in late 2023. In 2021, peaks were observed in February (2600 cases) and August (4100 cases), with the highest peak occurring between October and November, reaching approximately 4700 cases. In 2022, cases peaked in February (4300 cases), April–May (5500 cases, the highest), and July–September (5300 cases). The highest malaria burden was recorded in 2023, with a peak of 4900 cases in September and another between July and August, reaching 4700 cases.

3.3. Machine learning model performance

Using a four-fold cross-validation approach, the models were trained and validated iteratively across the folds. For each fold, performance metrics (MSE, MAE, and RMSE) were computed, and the mean values with their corresponding standard deviations were derived across the four folds. The comparative evaluation of model performance is summarized in Table 3.

From the results presented in Table 3, the Random Forest outperformed the other models, achieving the lowest average errors (MAE = 452.60 ± 431.20 ; MSE = $835,073.42 \pm 1,239,325.29$; RMSE = 675.52 ± 615.41), although the relatively large standard deviations suggest variability in its predictive accuracy across folds. Given the Random Forest performance, the model's predictive ability was visualised as presented in Fig. 6, comparing it with the test dataset of the actual malaria incidence.

3.4. Model interpretation and error diagnostics

3.4.1. Decision tree model

Fig. 7a and b and c presents the result of the performance of the decision tree model with 7a and b showing the distribution of the residuals and c showing the SHAP values and feature importance of the independent variable. The most contributing features are shown and ranked in decreasing order of mean absolute SHAP value (Fig. 7c). Horizontal violin plots (Fig. 7c) illustrate the distribution of feature impacts across all individuals in the test set. For each feature, relative values are represented by a colour gradient, with pink indicating higher values and blue indicating lower values.

The results indicate that Region (encoded) was the strongest predictor of malaria incidence, followed by LGA_encoded PeriodOrdinal (time). Climatic variables (Temperature and Rainfall) had smaller but measurable contributions.

3.4.2. XGBoost and Random Forest models

The performance of the XGBoost model is presented in Fig. 8a, b, and c. Fig. 8a and b displays the distribution of residuals, while Fig. 8c shows the SHAP values and feature importance of the independent variables. The SHAP analysis indicates that geographic variables (Region and LGA) exert the strongest influence on malaria incidence predictions, followed by temporal variation (Period). Climatic factors (temperature and rainfall) contributed comparatively less to the model. Fig. 9 presents the results of the Random Forest model, which also identified Region

(Fig. 9c) as the most influential variable.

4. Discussion

This study aimed to assess the influence of climatic factors on malaria incidence across the diverse ecological zones of Taraba State. An exploratory analysis was conducted to examine malaria incidence patterns across the ecological zones. The findings revealed that the Montane ecological zone had the highest malaria incidence, with Sardauna and Bali LGAs contributing the most cases. Further analysis confirmed a bimodal distribution of incidence in the Montane zone using the dip test ($D = 0.083$, $p < 0.05$). This observation is in line with previous studies that have shown higher malaria transmission rates in elevated, humid regions with persistent rainfall, where ecological conditions are favourable for mosquito breeding.^{11,12} A significant proportion of this burden appears to come from Sardauna and Bali Local Government Areas (LGAs), located within the Montane zone. Although our study did not include agricultural data, it is important to note that these LGAs are known for extensive agricultural activities, particularly rice and tea farming, both of which require intensive irrigation.³⁸ Sardauna, in particular, houses the largest tea plantation in Africa, and such agricultural practices have been reported in the literature to generate man-made breeding sites for mosquitoes through irrigation channels and standing water.³⁹ Future studies that incorporate agricultural and land-use data could provide more direct evidence of this relationship and strengthen the understanding of how human activities interact with ecological conditions to influence malaria transmission. Time series analysis further showed that rainfall exhibited a positive correlation with malaria incidence across all ecological zones, whereas temperature was negatively correlated. Notably, the Northern Savannah zone showed the strongest negative correlation between temperature and malaria incidence (-0.496), while the Montane zone had the strongest positive correlation between rainfall and malaria incidence (0.362). These findings are consistent with earlier reports that identify rainfall as a key factor in enhancing mosquito vector habitats, especially in regions with limited drainage infrastructure.¹² Similarly, the observed inverse relationship between high temperature and malaria incidence aligns with the literature, suggesting that excessive heat may reduce mosquito survival and inhibit parasite development.¹⁴

Although the exploratory analyses provided useful initial insights, they could not fully capture non-linear relationships or quantify the relative importance of multiple predictors. Machine learning models were therefore applied to address these limitations. To address this gap, machine learning models were applied to further investigate the contribution of ecological, geographic, and climatic factors to malaria incidence, with SHAP analysis used to enhance the interpretability of the model outputs. The models revealed that ecological zones (regions) were the strongest predictors of malaria incidence across all the models, highlighting the central role of geographic heterogeneity in shaping transmission. SHAP analysis consistently identified “Region” as the most influential predictor across models, though the magnitude of its contribution varied. In Random Forest, the mean absolute SHAP value for the region was 805.56, in the Decision Tree model, it is 772.47, while in the XGBoost model, it is 932.77. The strong predictive contribution of ecological zone may reflect the fact that zonal classification captures several operationally relevant determinants of malaria risk that extend beyond monthly rainfall and temperature alone. These may include altitude, vegetation type, drainage patterns, agricultural water use,

Table 3
Performance metrics of the machine learning models considered.

Model	MAE (mean $\pm$ SD)	MSE (mean $\pm$ SD)	RMSE (mean $\pm$ SD)
Decision Tree	882.92 $\pm$ 233.89	1800509.56 $\pm$ 886729.71	1297.04 $\pm$ 343.78
Random Forest	452.60 $\pm$ 431.20	835073.42 $\pm$ 1239325.29	675.52 $\pm$ 615.41
XGBoost	730.18 $\pm$ 162.50	1329043.12 $\pm$ 758577.39	1109.55 $\pm$ 312.95

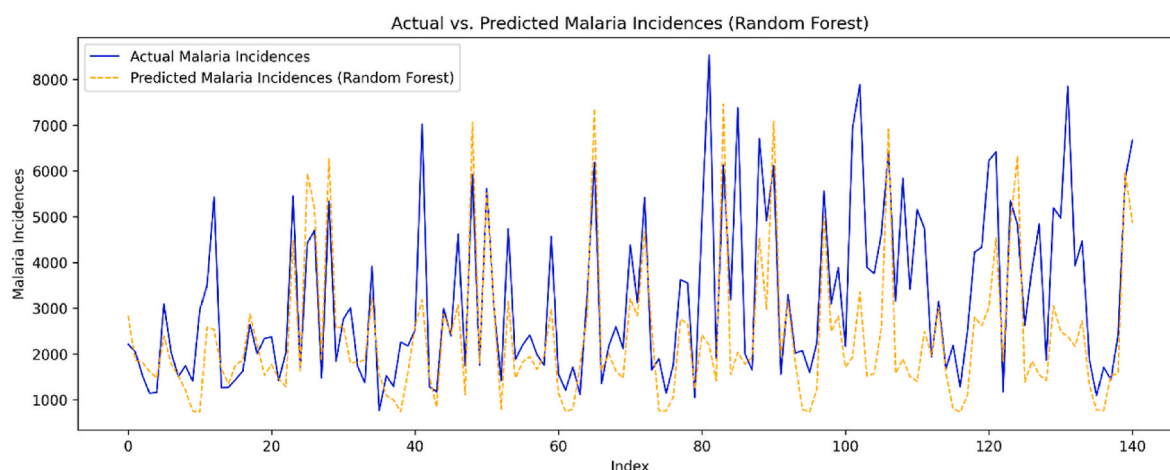


Fig. 6. Fitting curve of the Random Forest model on the actual malaria data set.

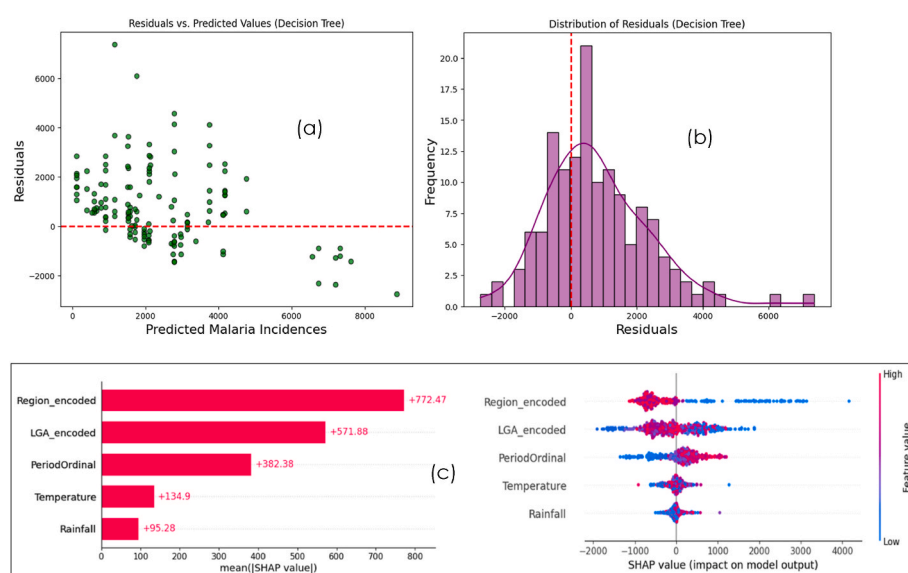


Fig. 7. Results of Decision Tree model (a) Residual plot (b) Distribution of the residuals (c) The Shapley (SHAP) values and feature importance.

housing structure, road access, health-service coverage, and differences in intervention reach across Taraba State.^{19,40} For example, the Montane areas of Sardauna and the surrounding Mambilla Plateau have distinct topography, cooler temperatures, and dispersed settlements that may influence both vector ecology and access to malaria prevention and treatment services.^{19,41} In contrast, savannah zones may differ in seasonality, settlement density, and health facility accessibility. This finding aligns with Nigeria's malaria control approach, which increasingly emphasises subnational stratification and context-specific planning under the National Malaria Strategic Plan.⁴² In practical terms, higher-burden zones identified through ecological profiling could be prioritised for intensified seasonal commodity pre-positioning, enhanced community case management, targeted long-lasting insecticidal net replacement campaigns, and stronger surveillance during peak transmission periods. Hard-to-reach highland or sparsely populated areas may additionally require adapted delivery models such as outreach services and microplanning to sustain intervention coverage. Lower-burden zones may benefit from surveillance-driven targeting and maintenance of routine prevention services. Incorporating ecological zoning into state malaria planning may therefore help improve allocative efficiency and operational precision in Taraba State. Random Forest demonstrated superior overall predictive performance based on

four-fold cross-validation across all tested metrics. This indicates that Random Forest not only captured the strong influence of ecological zones but also provided more reliable and balanced predictions compared to the other models.

These results underscore the growing body of evidence supporting the application of ensemble learning methods for health outcome prediction, particularly in settings with complex environmental dynamics.^{6,25}

Given these findings, malaria control strategies should consider the human practices that inadvertently support vector proliferation, especially in Sardauna and Bali LGA.

These tools can be instrumental in guiding timely and targeted interventions, such as the distribution of long-lasting insecticidal nets (LLIN) or public awareness campaigns.

While this study demonstrates the potential of machine learning in predicting malaria incidence, it is important to acknowledge certain limitations. The reliability of climate-based malaria prediction largely depends on the quality and consistency of the underlying data. In this case, malaria case data were sourced from DHIS2, which is prone to variations in reporting practices across LGAs, as well as the exclusion of private facilities. Some LGAs may report more completely or frequently than others, leading to potential underreporting or overreporting.

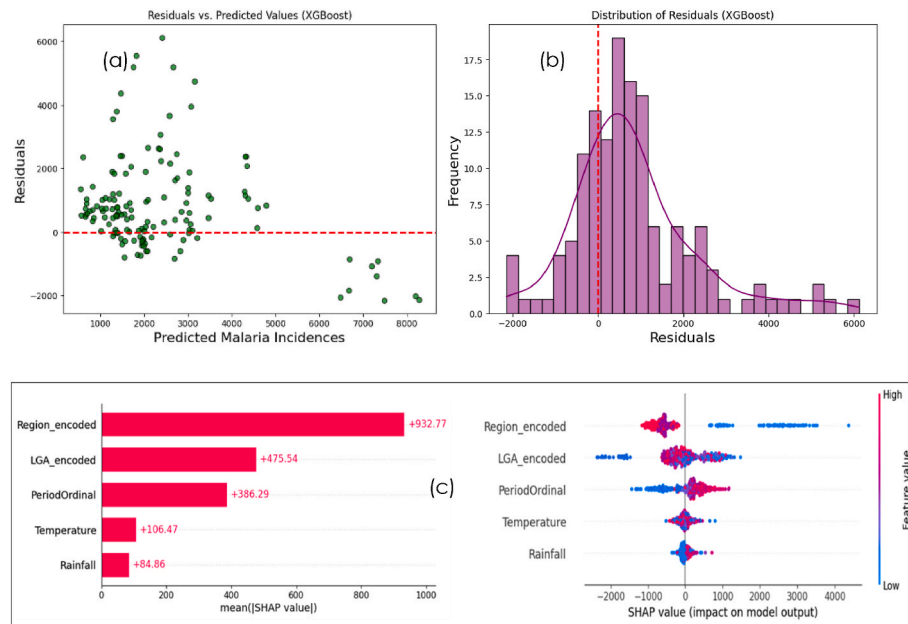


Fig. 8. Results of XGBoost model (a) Residual plot (b) Distribution of the residuals (c) The Shapley (SHAP) values and feature importance.

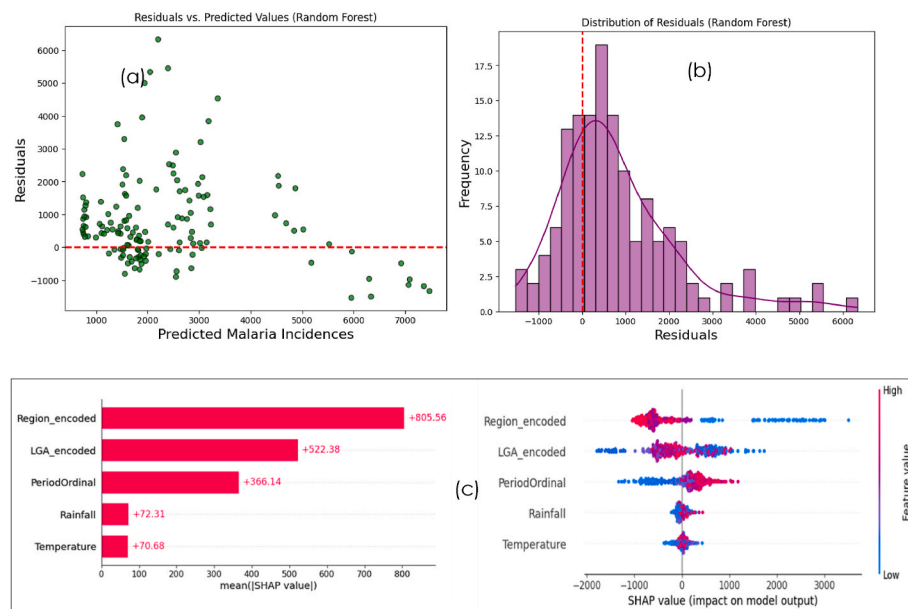


Fig. 9. Results of Random forest model (a) Residual plot (b) Distribution of the residuals (c) The Shapley (SHAP) values and feature importance.

DHIS2 records show that health facility attendance in Taraba State recorded a decline from 1,217,147 to 1,081,693 (−11.1%). This reduction suggests potential underutilization or reporting gaps during the pandemic, consistent with Nigerian studies documenting COVID-19–related disruptions in health service delivery.³⁰ As a result, the data may not fully capture the true malaria burden across the study area, which could impact the accuracy of the predictive models. Secondly, this study primarily focused on climatic and ecological variables without incorporating factors such as *Anopheles* mosquito vector dynamics, human behavioural patterns, drug resistance, and intervention coverage. This was due to the lack of reliable, zone-level data on entomological and socio-behavioural factors in Taraba State. Although climate variability is a well-established driver of malaria seasonality, the absence of these additional determinants may restrict the comprehensiveness of the models. Also, the findings are region-specific to Taraba

State and may not be directly generalizable to areas with different contexts. The ecological diversity within Taraba offers a valuable test-bed for studying malaria–climate interactions, but caution should be taken when extrapolating these results to other regions.

Despite these limitations, this study contributes valuable insights into the climate–malaria relationship in a highly heterogeneous ecological setting and demonstrates the utility of machine learning methods for both prediction and feature importance analysis. Future studies should seek to incorporate entomological, behavioural, and intervention-related variables, and apply the framework across multiple geographic regions to improve both generalizability and policy relevance.

5. Conclusion

This study employed machine learning approaches to assess the influence of climatic and ecological factors on malaria incidence in Taraba State, Nigeria. The Random Forest model achieved the best overall predictive performance among the models considered, indicating potential utility for capturing complex non-linear relationships in malaria data. Across models, ecological zone and geographic location emerged as important predictors, while rainfall and temperature showed additional contributions. Collectively, the findings suggest that malaria burden in Taraba State may vary according to both environmental and geographic context. Integrating such heterogeneity into surveillance systems and malaria control planning may enhance the targeting of locally appropriate interventions.

CRedit authorship contribution statement

Dupsy Akoma: Writing – original draft, Writing – original draft, Visualization, Software, Methodology, Formal analysis. **Yusuf Abdulkareem:** Software, Methodology, Formal analysis. **Ganiyat Eshikhena:** Project administration. **Obonganwan Akpabio:** Data curation. **Atinuke Arthur:** Writing – review & editing. **Zainab Abdullahi:** Data curation. **Nicholas Oyeh:** Data curation. **Abayomi Amire:** Data curation. **Christian Abba:** Data curation. **Charles Obi:** Formal analysis. **Gbenga Adegbite:** Writing – review & editing. **Adaeze Eche-George:** Data curation, Conceptualization. **Ifeoma Ezenyi:** Writing – review & editing. **Ezra Gayawan:** Writing – review & editing. **Eze Nelson:** Writing – review & editing. **Chijioke Kaduru:** Provided leadership of project design, supervision of project delivery, and supervisory authorship of our manuscript.

Ethics approval and consent to participate

Not applicable. This study did not involve the collection of primary data from human participants. It utilized secondary data on malaria cases obtained from the District Health Information System (DHIS2), with permission granted by the National Malaria Elimination Programme (NMEP), Nigeria.

Consent for publication

All authors have agreed to publish the paper.

Availability of data and materials

The malaria case data used in this study were obtained from the District Health Information System 2 (DHIS2) and was granted access by the National Malaria Elimination Programme (NMEP), Nigeria. The rainfall and temperature data were obtained from the Nigerian Meteorological Agency (NiMet). Access to both datasets is restricted and was granted specifically for the purposes of this study. Access to these datasets may be restricted due to institutional data-sharing policies; requests for access should be directed to the respective institutions (NMEP and NiMet).

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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List of abbreviations

XGB	Extreme Gradient Boosting
DHIS2	District Health Information System 2
LGA	Local Government Area
PW	regnant Women
RFR	Random Forest Regression
MSE	Mean Square Error
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
LLIN	long-lasting insecticidal nets
NMEP	National Malaria Elimination Programme
NiMet	Nigerian Meteorological Agency
NMMF	Nigeria Malaria Modelling Fellowship

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